

T9.2 – Decoding root apoplastic barrier formation and plasticity from single-cell identities to whole root architecture – MASLARD Corentin

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Root apoplastic diffusion barriers control the movement of water and nutrients, and play essential roles in plant development and environmental adaptation. In grasses, the exodermis forms a major outer root barrier, yet the molecular mechanisms underlying its formation, regulation, and integration with whole-root developmental plasticity remain incompletely understood. Here, a multiscale approach is presented to investigate root apoplastic barrier formation by combining single-nucleus RNA sequencing, reporter-based cell identity analysis, and AI-assisted root phenotyping. Single-nucleus transcriptomic datasets were used to resolve root cell populations associated with barrier development and to identify candidate molecular programs involved in hydrophobic polymer deposition, cell wall remodeling, and barrier differentiation. Barrier-associated clusters were characterized by specific expression of CASP-related markers, together with candidate genes from families such as GPATs, ABC transporters, peroxidases, and wall-remodeling enzymes, suggesting coordinated transcriptional programs underlying endodermal and exodermal differentiation. To connect these cellular signatures with root system-level phenotypes, an AI-based image analysis pipeline was established for automated root detection and quantitative root architecture analysis. This pipeline enables robust extraction of architectural traits from plant growth experiments and provides a scalable strategy to compare contrasting *Brachypodium* genotypes under abiotic stress conditions, including salt, ABA, and nutrient deficiency treatments. By linking cell-type-specific molecular signatures with quantitative root phenotypes, this work explores how barrier-associated programs translate into whole-root architectural responses. Together, this study bridges molecular cell identity, candidate barrier-regulatory pathways, and root system architecture, providing a framework to investigate how apoplastic diffusion barriers are formed and functionally connected to adaptive root growth in grasses.